CAD Design of an Adjustable Mobility Crutch

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Introduction

- The medical device I designed for my cad project was a height-adjustable axillary crutch meant to assist people with temporary or chronic lower-limb injuries. This design takes inspiration from the standard crutch, while making it height adjustable and dimensionally accurate via computer aided design (CAD) in OnShape.

Size and Justification

- The overall height (collapsed) of the crutch is 1023 mm, with there being a travel of around 180 mm, and thus the maximum height gets to ~1203 mm. We are using the 77% rule, that an axillary crutch length should be 77% of a person's height [1] . The target audience for this crutch are patients with a height of ~1.33–1.56 m, so pediatric and short adolescent. I chose this size and target audience mainly due to the fact that these types of crutches are used in great lengths by kids growing up, and this I wanted to cater my product to this audience.

List of Parts

Upper Frame

- This part was used twice. The left side of the frame was mirrored to create the right side,, so there are 2 parts on the onshape design.
- Forked shaft design that makes the bulk of the crutch, provides structure and a place to house the arm and hand support. It also allows a space in between the 2 frames to hold the lower frame that allows the crutch to be telescopic. There are pin holes in this to hold the hand support and to connect the upper frames to the lower frame.

Lower Frame

- This part was used once.
- Inner telescoping tube with pin holes for height adjustment.

Frame Connector

- This part was used once.
- Three-way joint linking the upper frame, the lower frame, and the telescopic leg.

Extendable Leg

- This part was used once.
- Inner telescoping tube that slides within the frame to adjust crutch height. Includes locking holes for secure positioning.

Cap

- This part was used once.
- End plug that seals the lower tube, preventing debris entry and provides a grip to the floor.

Shoulder Rest

- This part was used once.
- Contoured top pad that distributes underarm load evenly for user comfort.

Hand Rest

- This part was used once.
- Cylindrical grip for hand placement; supports body weight and maintains stability.

Peg Extendable

- This part was used once.
- Locking pin that engages holes in the telescoping leg to fix height.

Peg Frame Connection

- This part was used once.
- Pin that secures the connector body to the main frame structure.

Peg Hand Connection

- This part was used once.
- Fastening peg that connects the hand rest to the central connector, allowing easy removal or replacement.

Upper Frame

Pictures

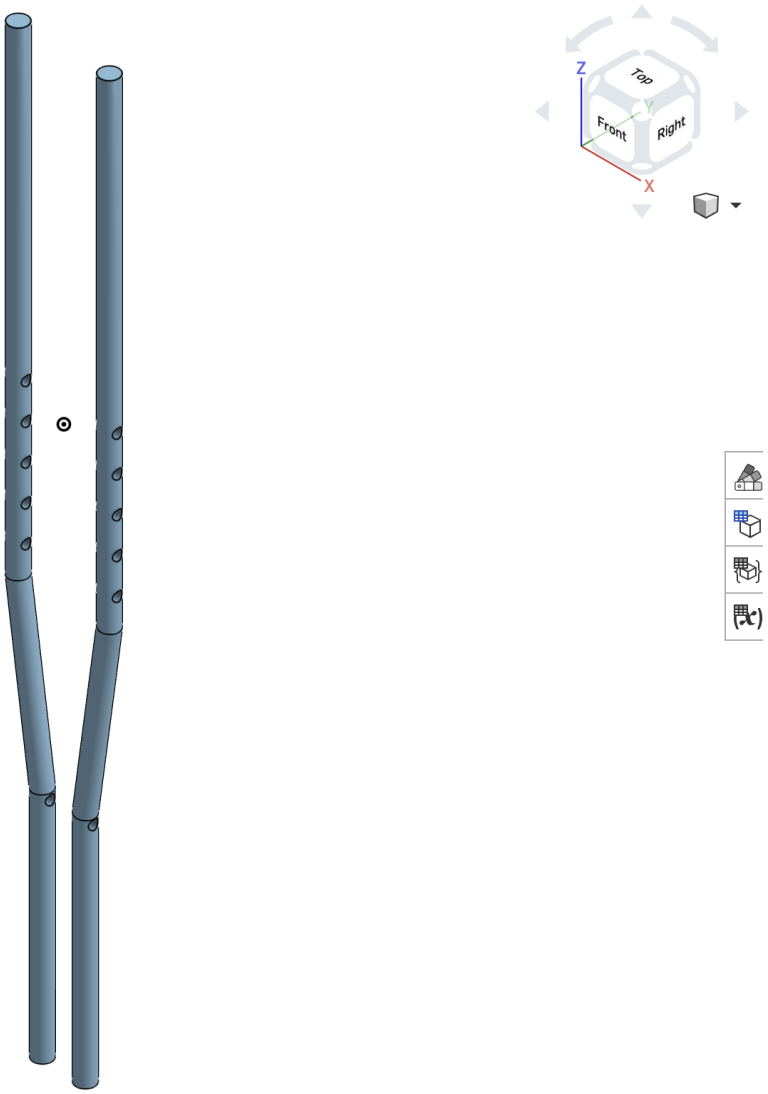


Figure 1: Upper Frame from a standard isometric view

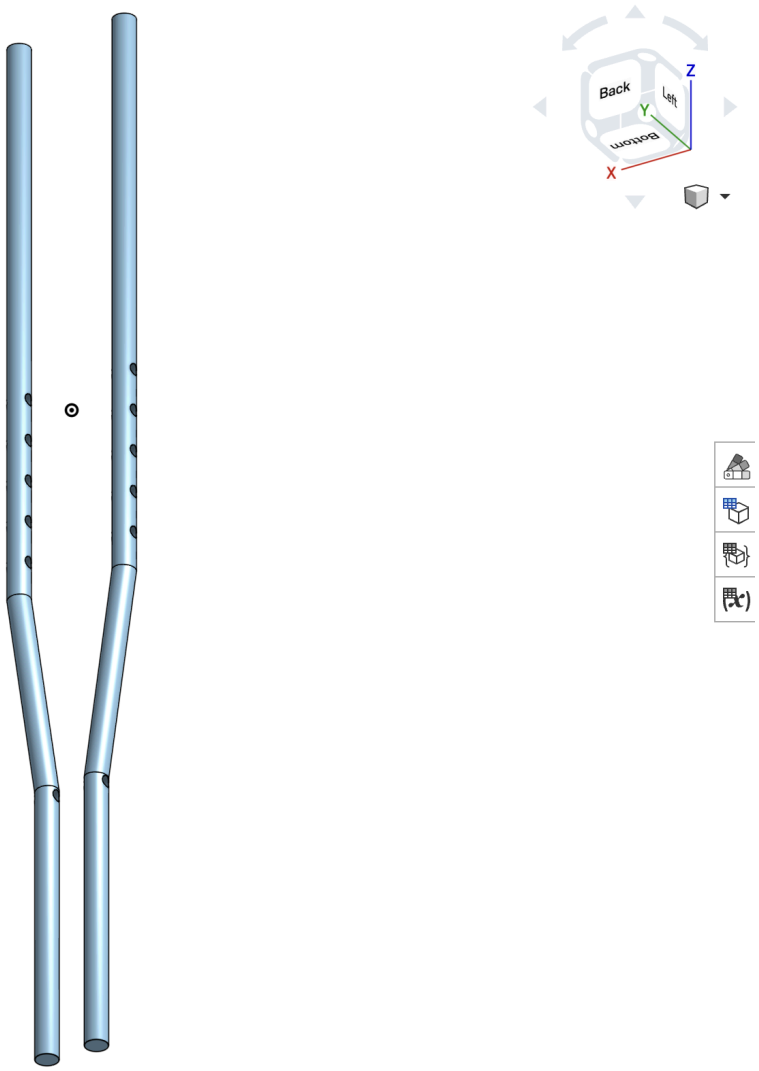


Figure 2: View of Upper Frame shown from the bottom and back. These pin holes are where the hand rest (top 5 pins) and the pin to connect all 3 frames (bottom 1 pin) attach to.

Multiview drawing

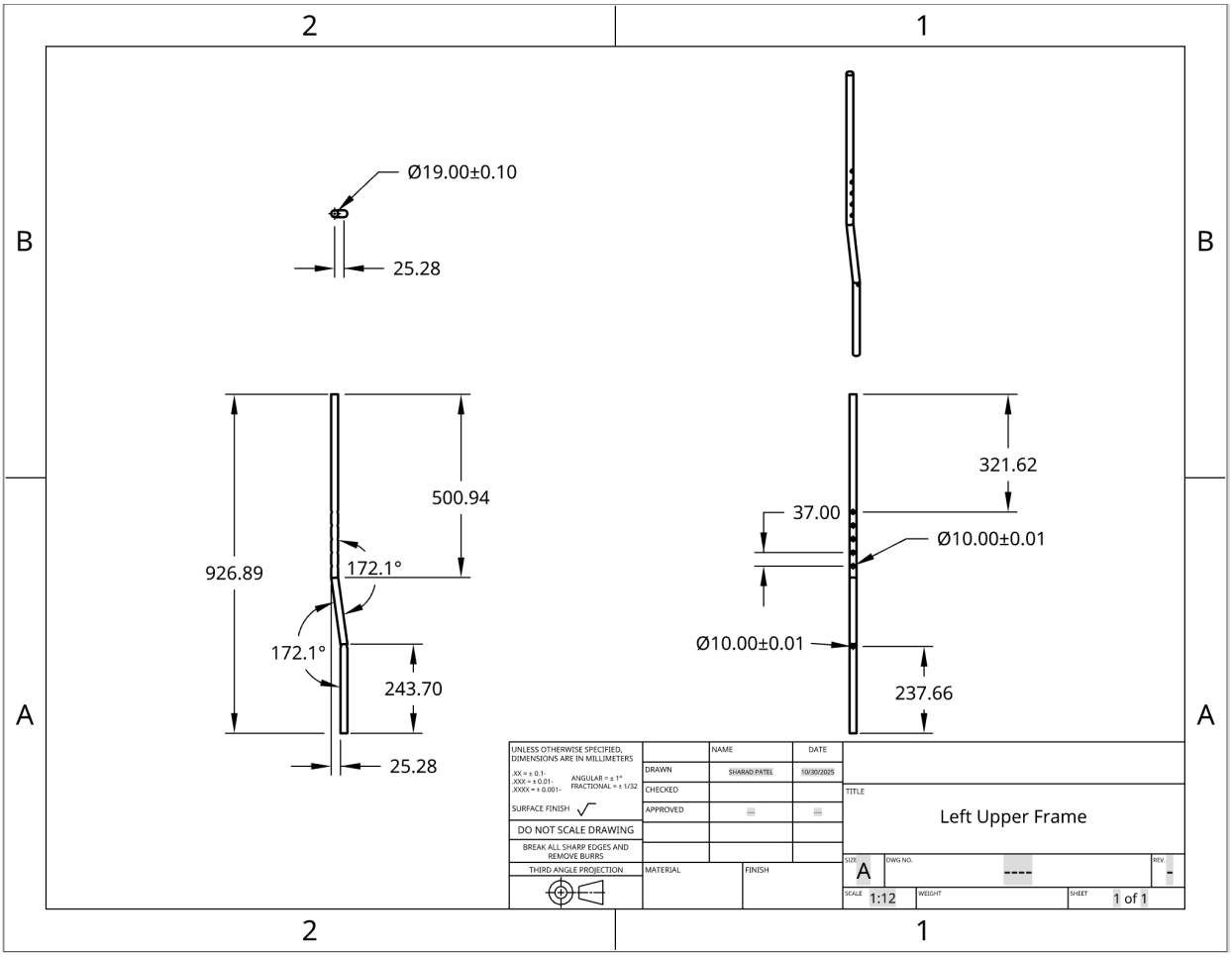


Figure 3: Multiview drawing of the left upper frame

Notes

I feel like the materials that best work for this is an aluminium alloy, as that has the best balances of strength, weight, and corrosion resistance. For all the drawings, I have general tolerances listed at the bottom. For the general linear tolerances (.XX = ±0.1, .XXX = ±0.01, .XXXX = ±0.001), this allows for a good intersection between manufacturability and function. The angular and fractional tolerances are not too important to this design, so those tolerances were left board.

The main tolerances I specified were the ones where 2 parts were interacting/sliding with each other. For this, I put a tolerance of 0.01 for all the peg hoes in my design. These holes do not need to be so specific, however as the holes were small already, having a too large tolerance might affect how the peg and the holes interact, especially since the peg has to cross multiple holes to get to the other side (one for the left frame, one for the right).

Lower Frame

Pictures

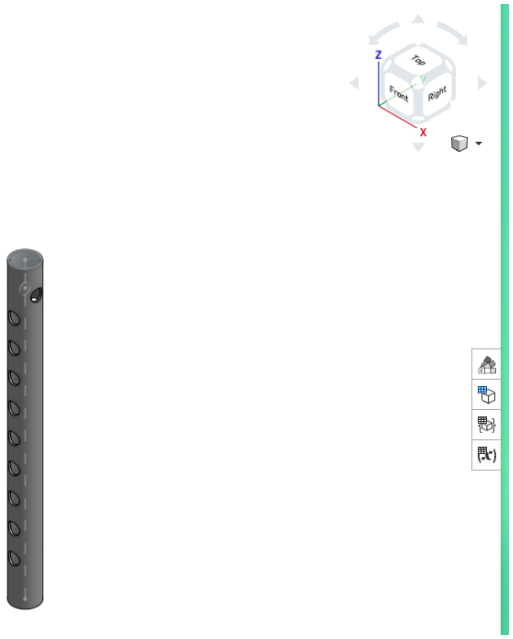


Figure 4: Standard isometric view of lower frame

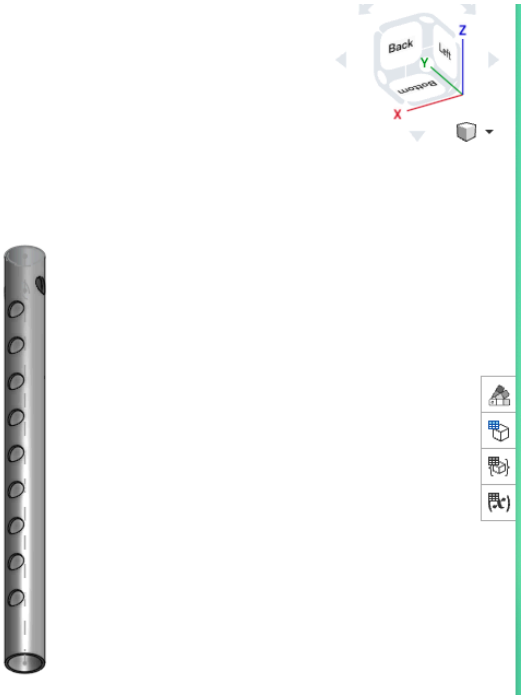


Figure 5: View of lower frame from back-left angle. The pin holes are were the 3 frames are attached together (top pin hole) and where the extendable leg can connect (bottom pin hole)

Multiview drawing

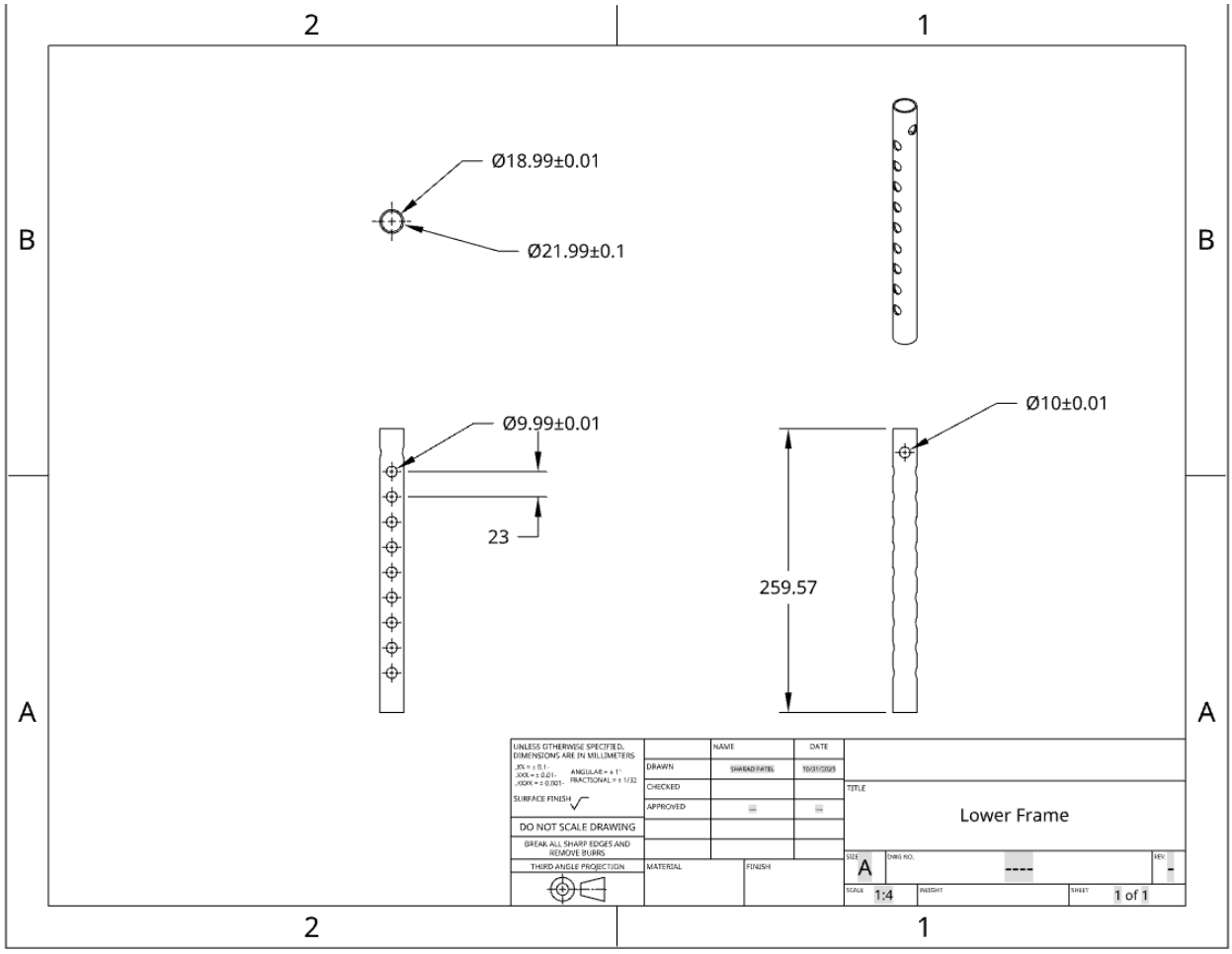


Figure 6: Multiview drawing of the lower frame

Notes

The ideal material for this will be aluminium alloy, the same as the upper frame for the same reasons. The tolerances are based of the standard linear tolerances provided below, with the pin holes having the same tolerance as the pin holes for the upper frames. The tolerance for the inner circle of the rod is also 0.01, as the extendable leg will be slide though this, and this you need a tight tolerance.

Frame Connector

Pictures

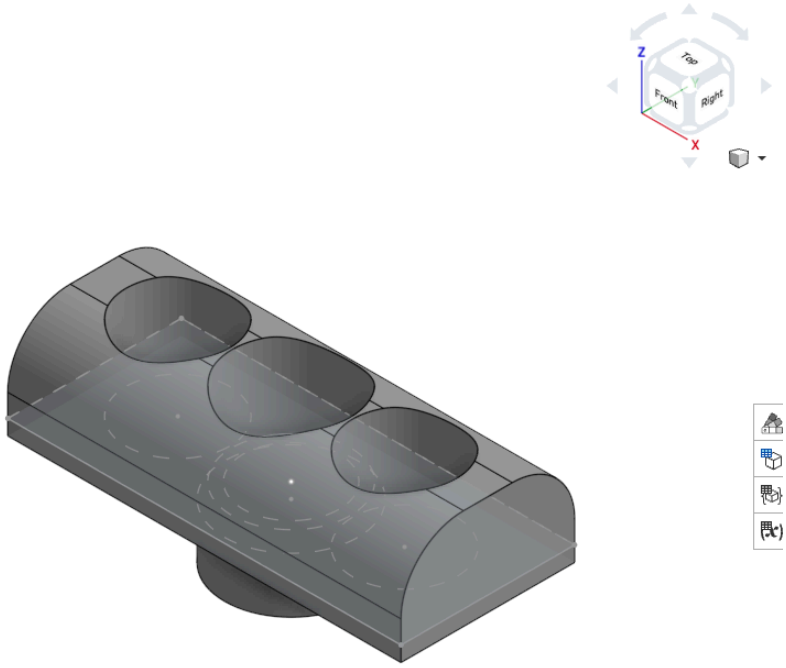


Figure 7: Standard isometric view of the frame connector

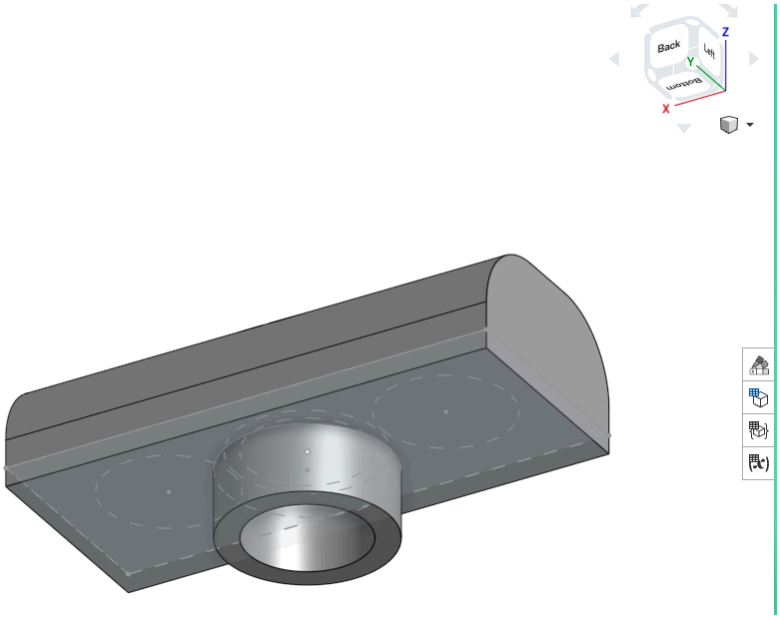


Figure 8: View of frame connector from back-left angle. The 3 holes on the top are for the 2 upper frames and the 1 lower frame, while the hole in the bottom is for the extendable leg.

Multiview drawing

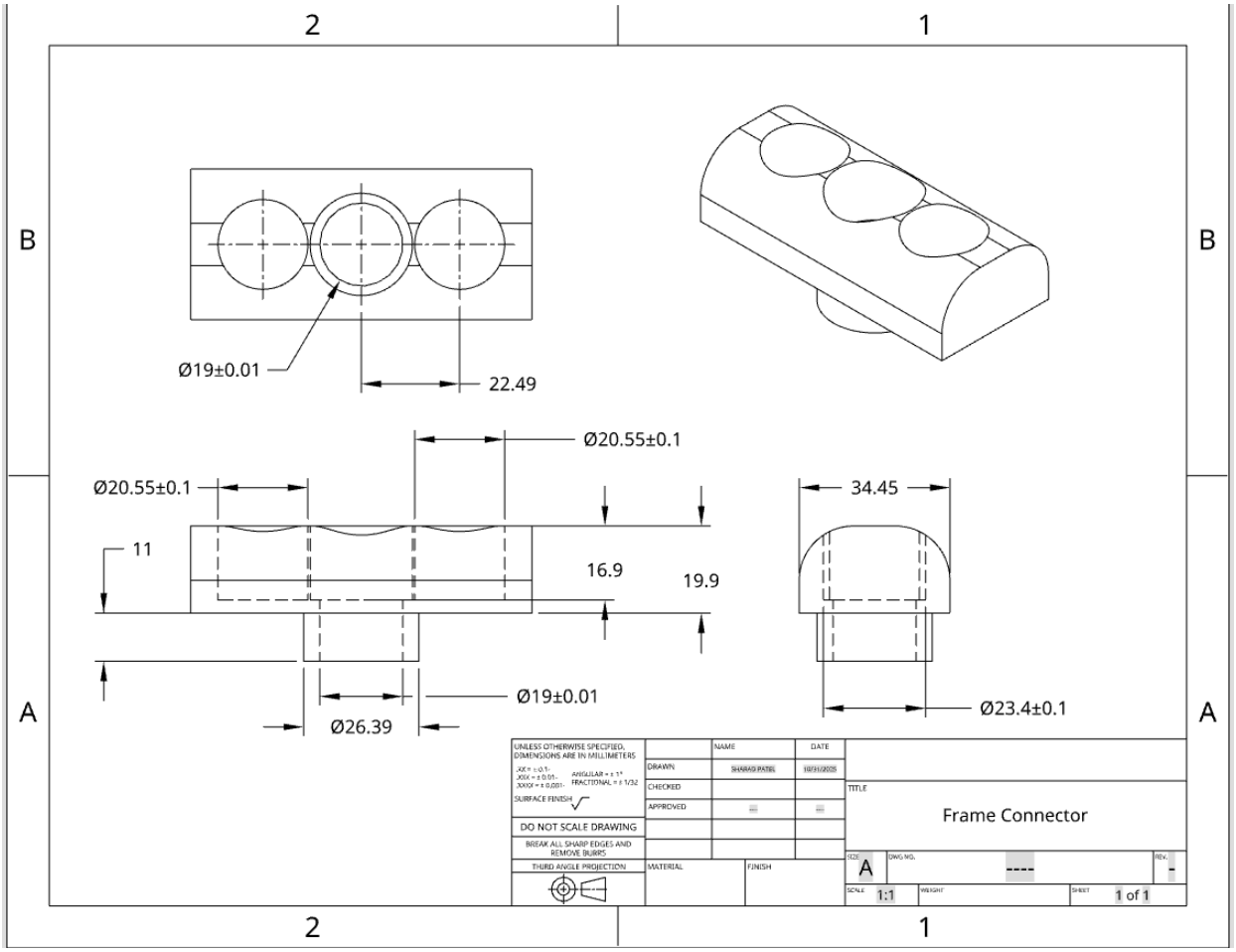


Figure 9: Multiview drawing of the Frame Connector

Notes

The best material for this is a polymer, ideally ABS, ideally for the ease of printing/molding and the weight. For this, we use the same generic tolerances for the reasons provided above. In this, we are connecting the 3 frame rods into the top of this connector. Since there is no movement and the rods can be attached via other methods (ie. Glue, welding), those dimensions are not too tight, at 0.1 mm. The main focus was the bottom hole. This is where the extendable leg will be moving up and down, and this we have a tolerance of 0.01 to make a clean and error free outcome.

Extendable Leg

Pictures



Figure 10: Standard isometric view of the the extendable leg



Figure 11: View of extendable leg from back-left angle. The peg hole is to make the crutch adjustable.

Multiview drawing

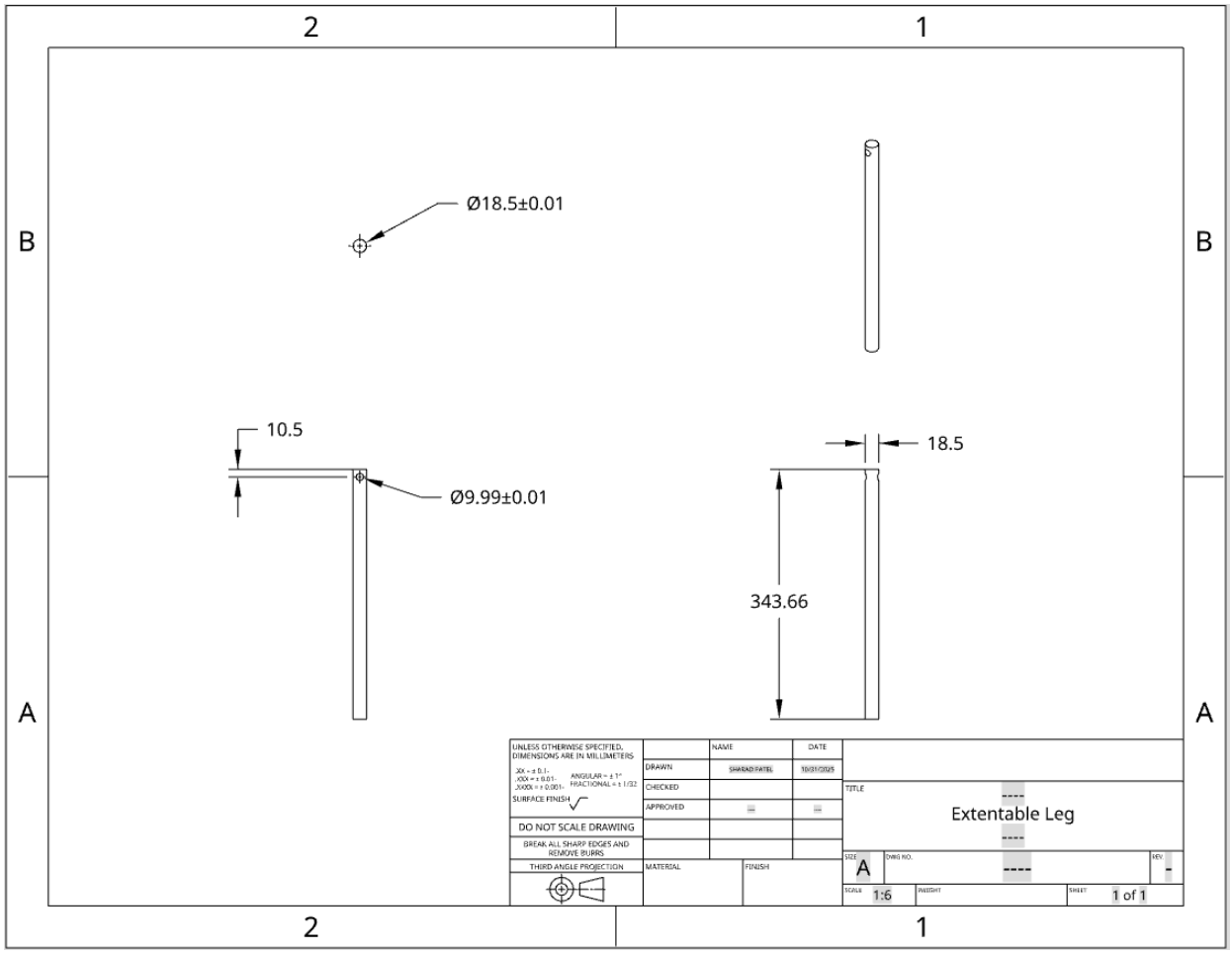


Figure 12: Multiview drawing of the extendable leg

Notes

Similarly to the frames, the best material to make this out of is aluminum alloy. The general tolerances apply to this part. The main specifications here are the outer diameter of leg and the peg hole. Both of these are parts where it will be sliding into another thing or a peg is sliding through it, and this the tolerance should be small to get the best prints.

Cap

Pictures

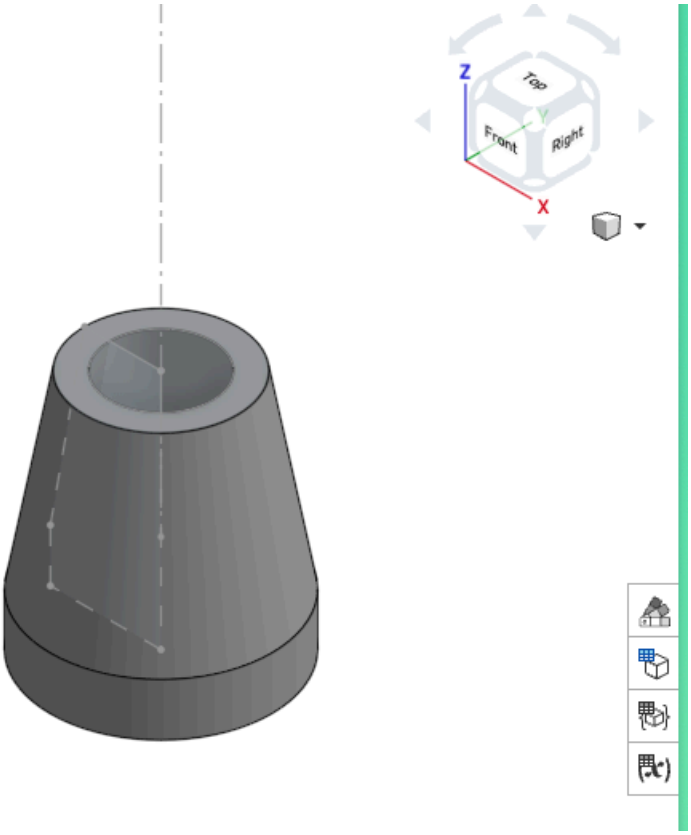


Figure 13: Standard isometric view of the the cap

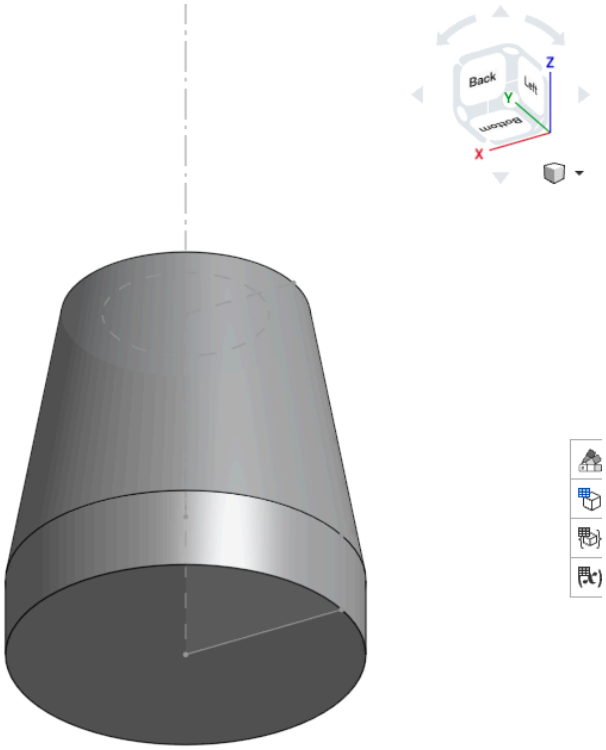


Figure 14: View of cap from back-left angle. The hole on top is to fit the extendable leg into it

Multiview drawing

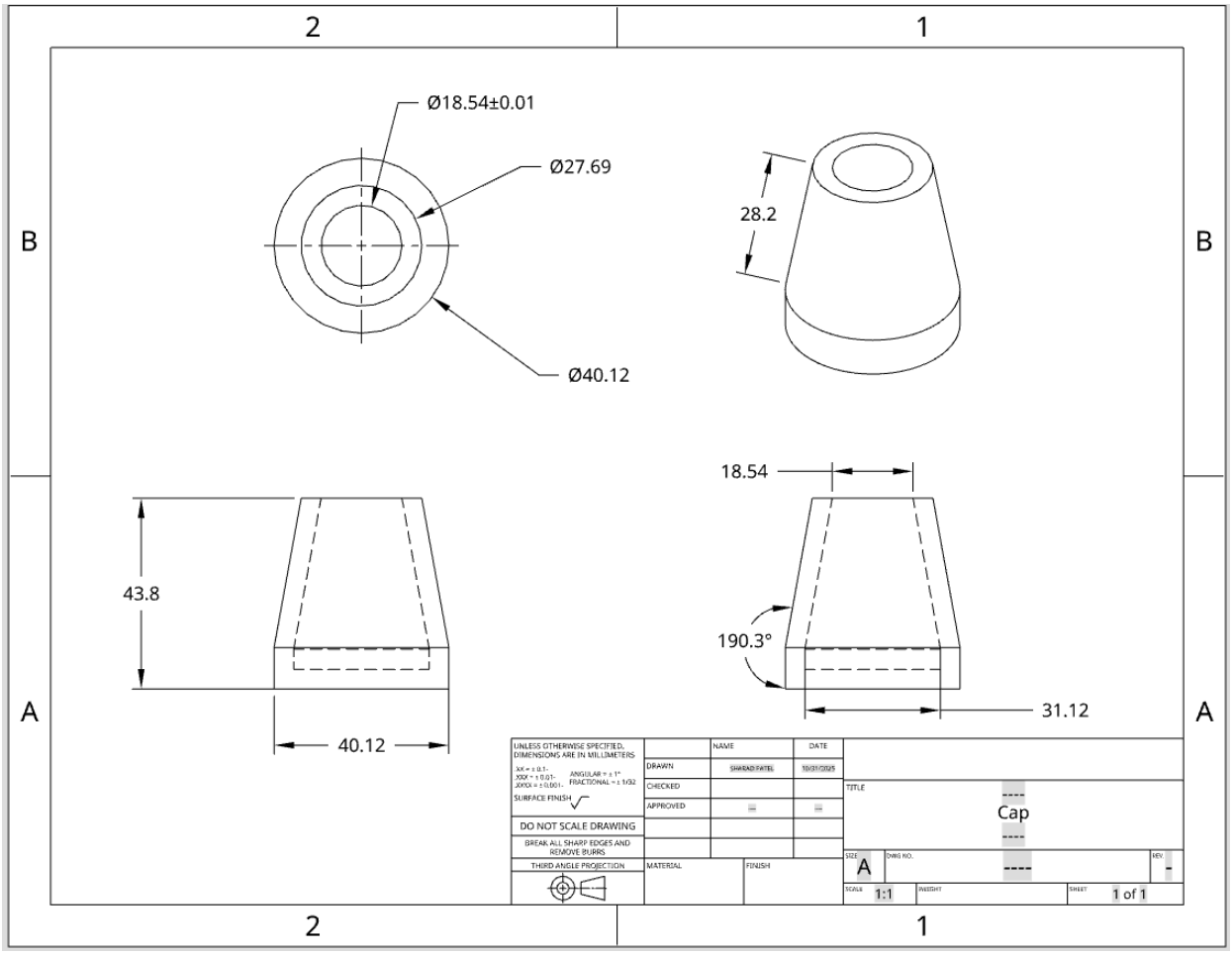


Figure 15: Multiview drawing of the Cap

Notes

The best materials for this cap part is a rubber mold. That will be the best to have the most contact and grip to the group while using the crutch. The only main tolerance outside the general tolerances listed at the bottom is the inner diameter to be attached to the extendable leg. This needs to be tight fit and was designed as such. So a tolerance of 0.01 mm ensures there is no loose fit/

Hand Rest

Pictures

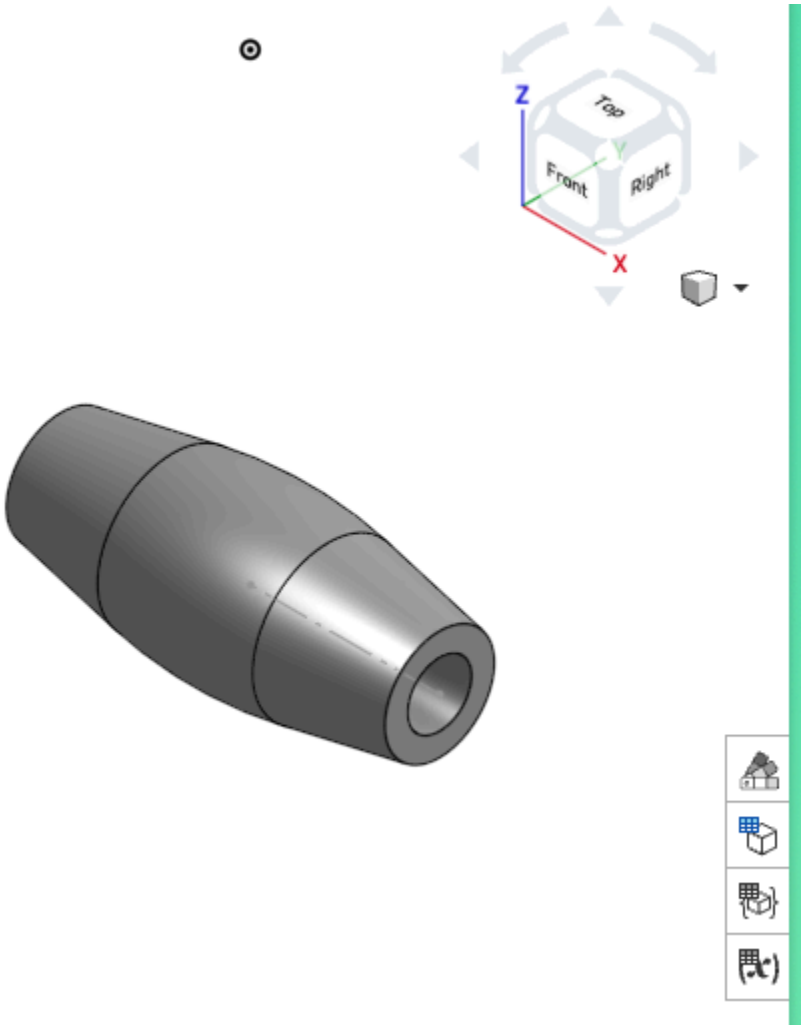


Figure 16: Standard isometric view of the the hand rest

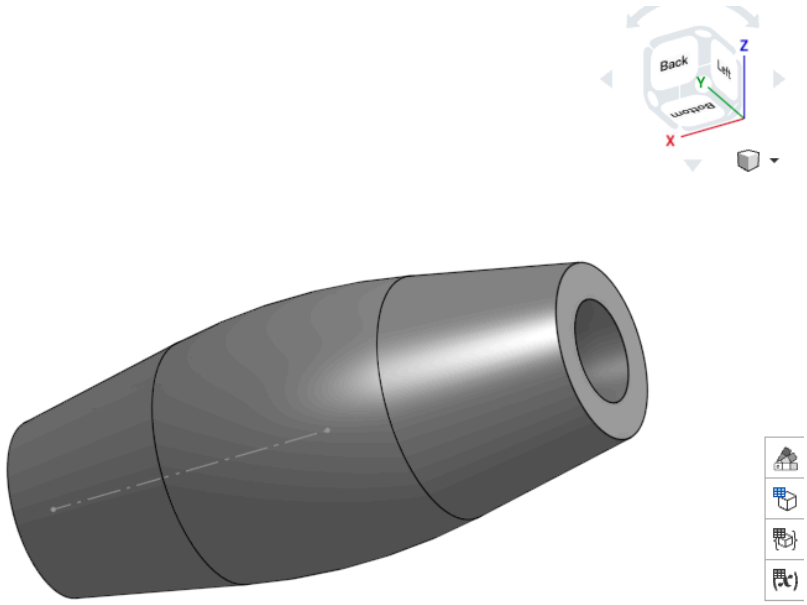


Figure 17: View of hand rest from back-left angle. The hole spanning the design is for a peg to fit this in between the 2 frames.

Multiview drawing

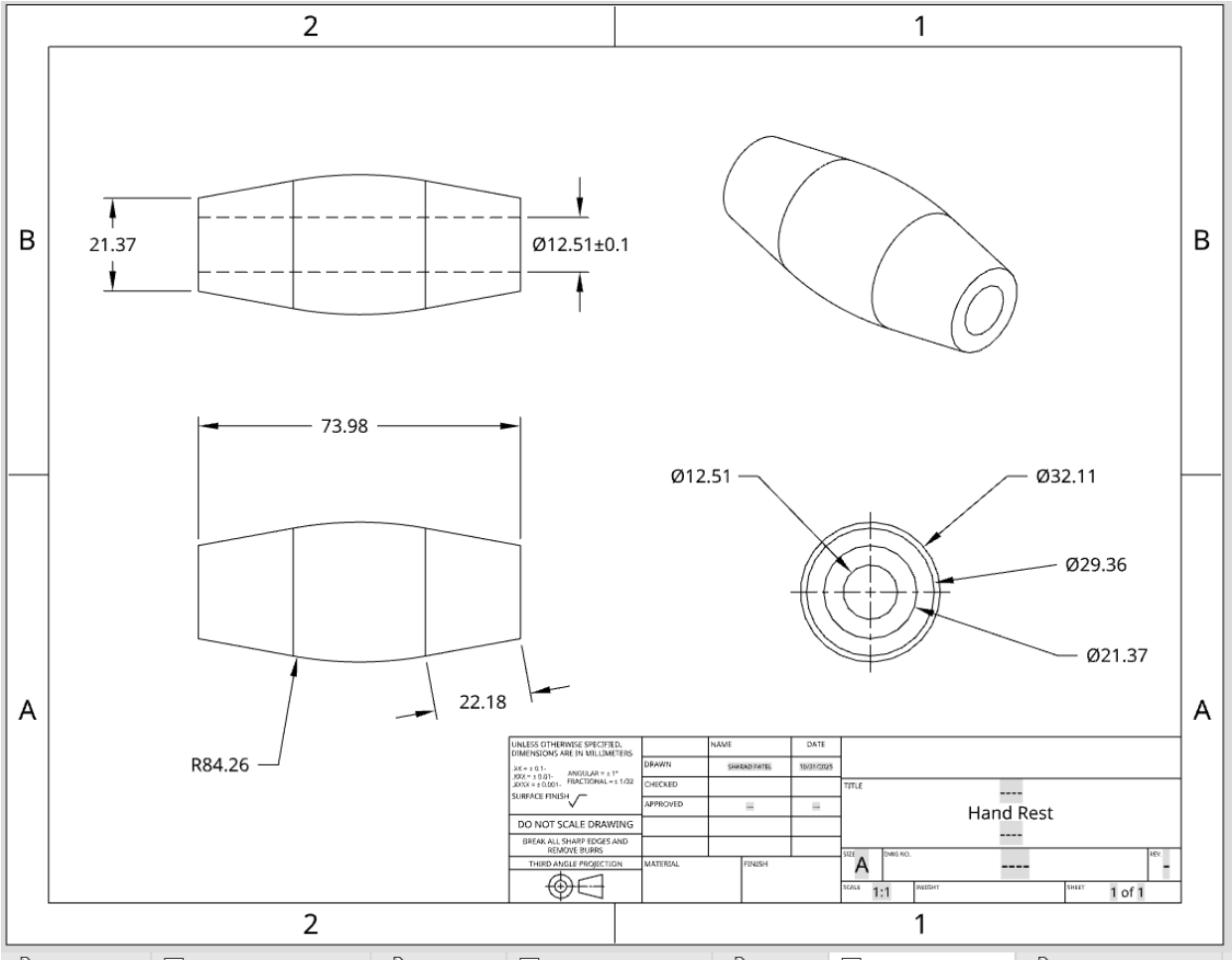


Figure 18: Multiview drawing of the Cap

Notes

A TPU or a rubber material will be the best for the best comfort and pressure relief for this piece. There are the general tolerances at the bottom, however the inner diameter for the circle is specific and needs to be 0.01 mm. Again, there isa pin going through this, and as a result, to ensure best accuracy, there must be a tight fit.

Shoulder Rest

Pictures

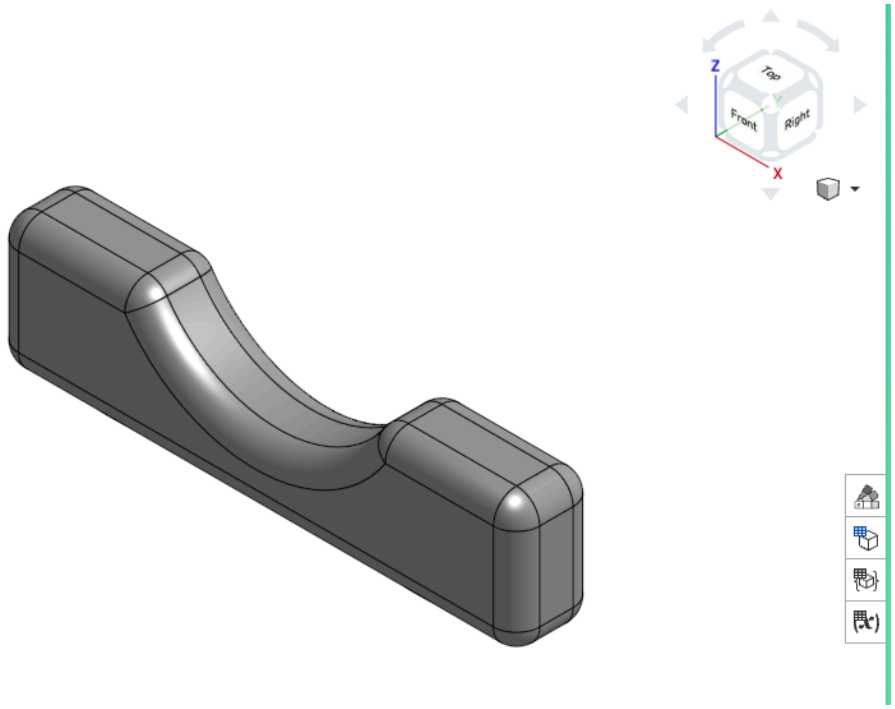


Figure 19: Standard isometric view of the the Shoulder Rest

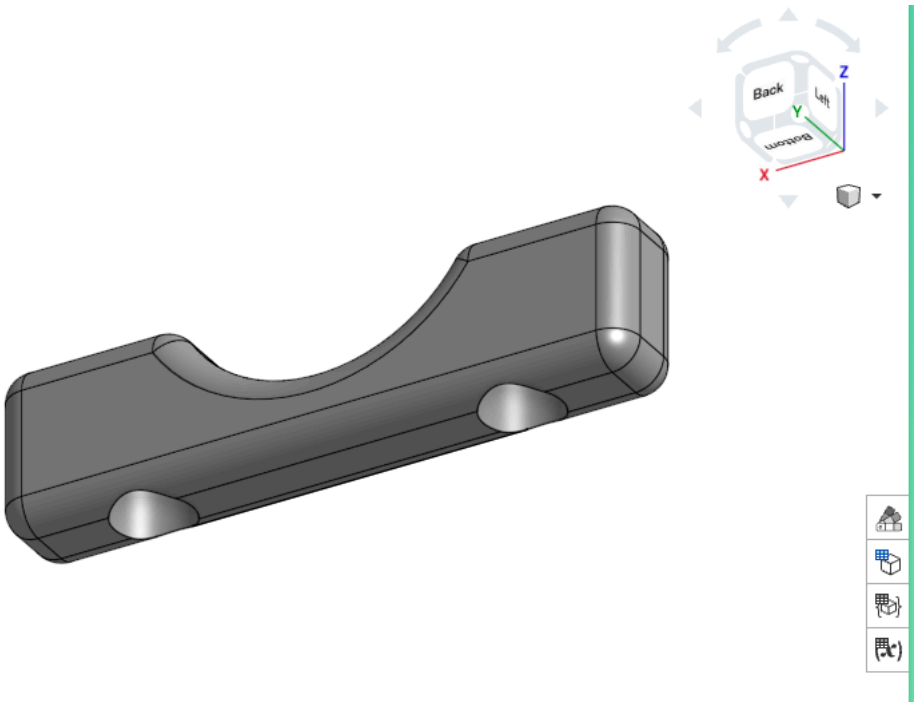


Figure 20: View of hand rest from back-left angle. The 2 holes at the bottom are to fit the upper frames into it.

Multiview drawing

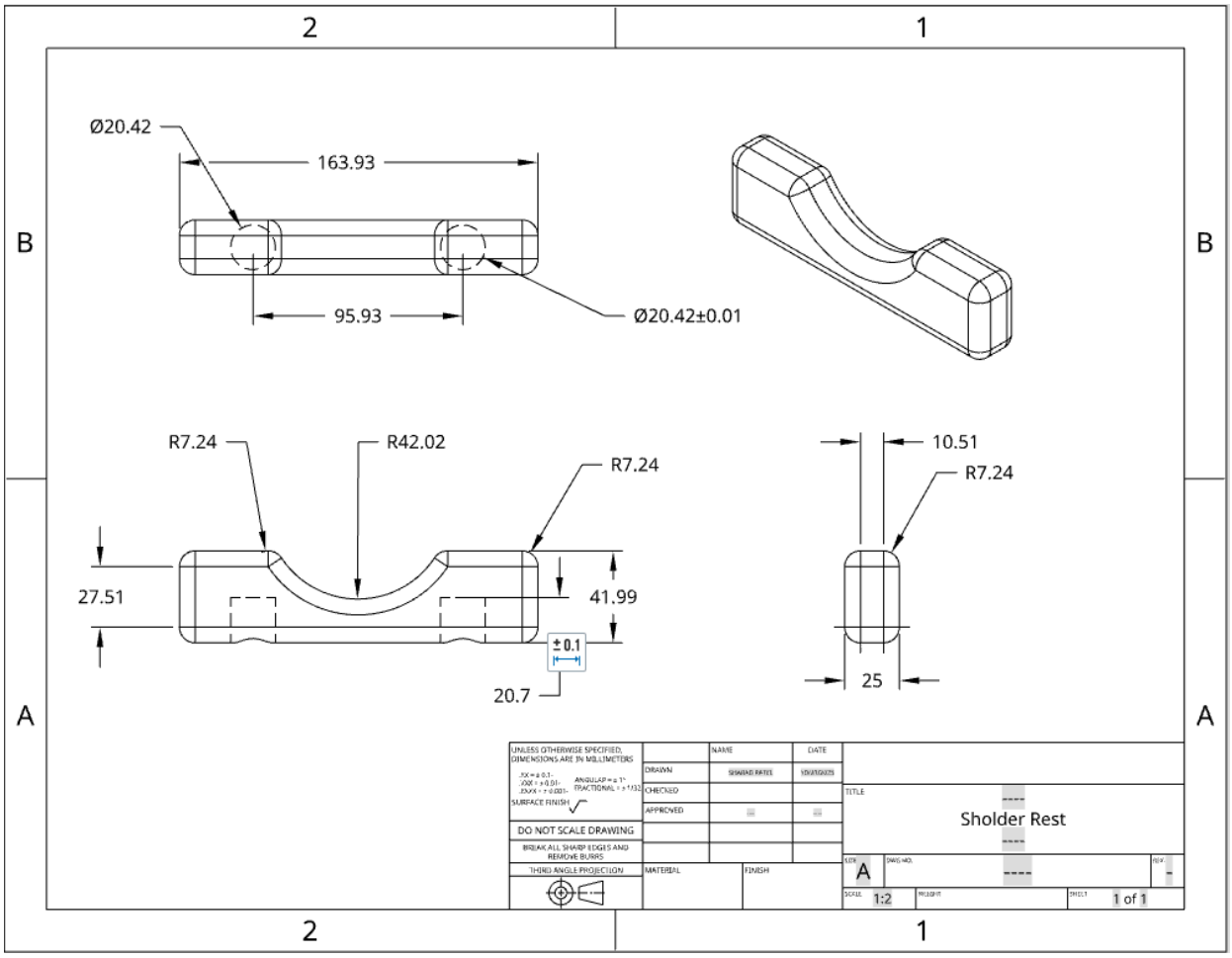


Figure 21: Multiview drawing of the Cap

Notes

This is very similar to the hand rest, where a plastic/polymer base with TPU is ideal for molding to the persons need without losing shape. The main tolerance that is specified outside the general tolerances are the holes were the upper frames go. This needs to be a tight fit to be proper, and thus, a tolerance of 0.01 is specified.

Pegs (Extenable Leg, Frame Connection, Hand Rest)

Pictures

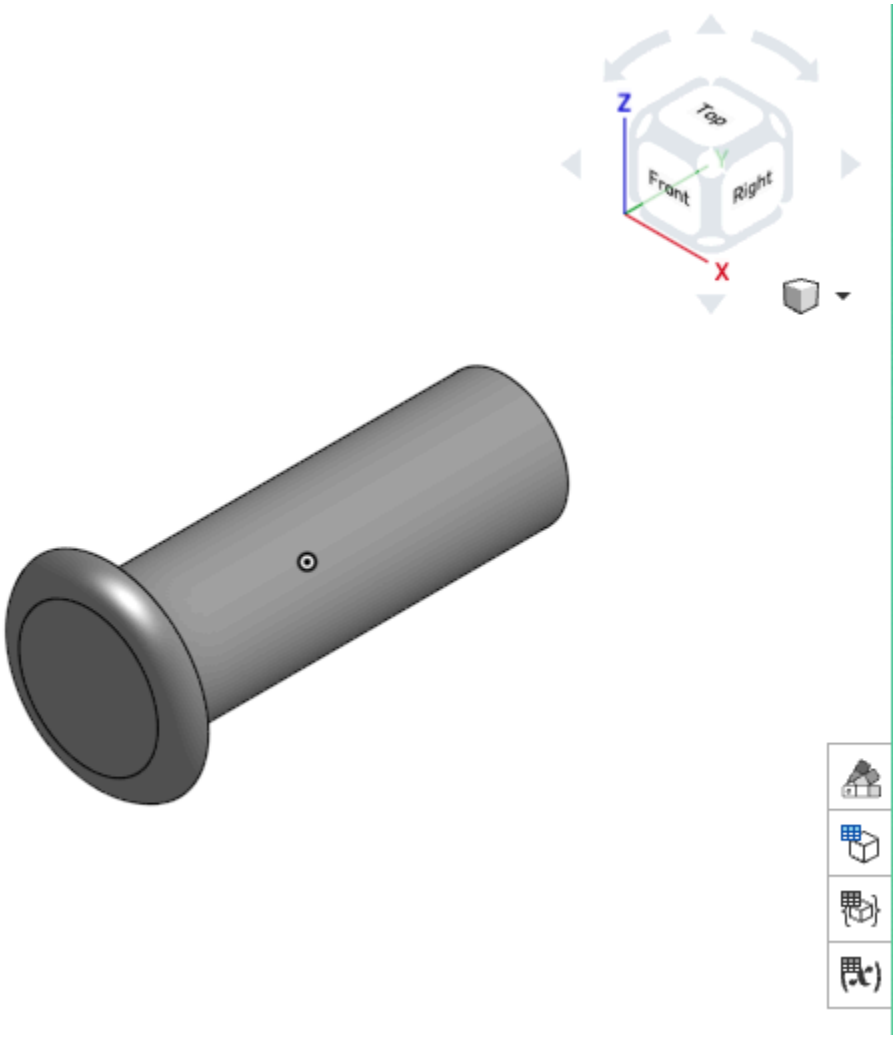


Figure 22: Standard isometric view of the Peg extentable

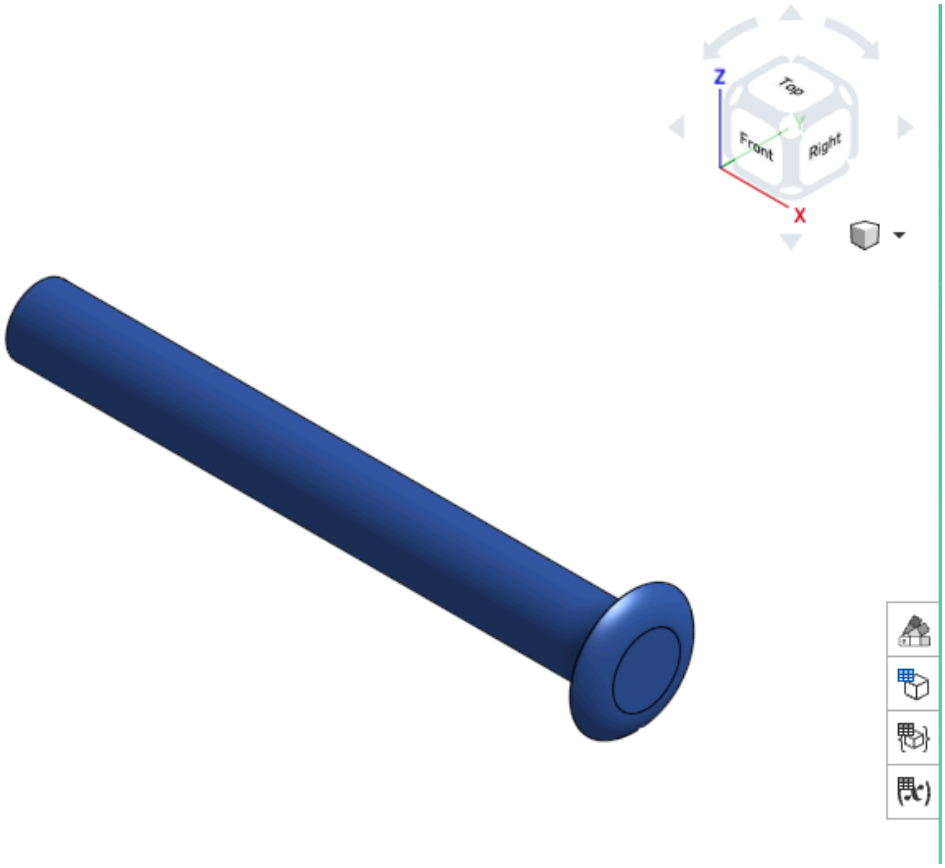


Figure 23: Standard isometric view of the Peg Frame connection



Figure 24: Standard isometric view of the Peg hand rest

Multiview drawing

was

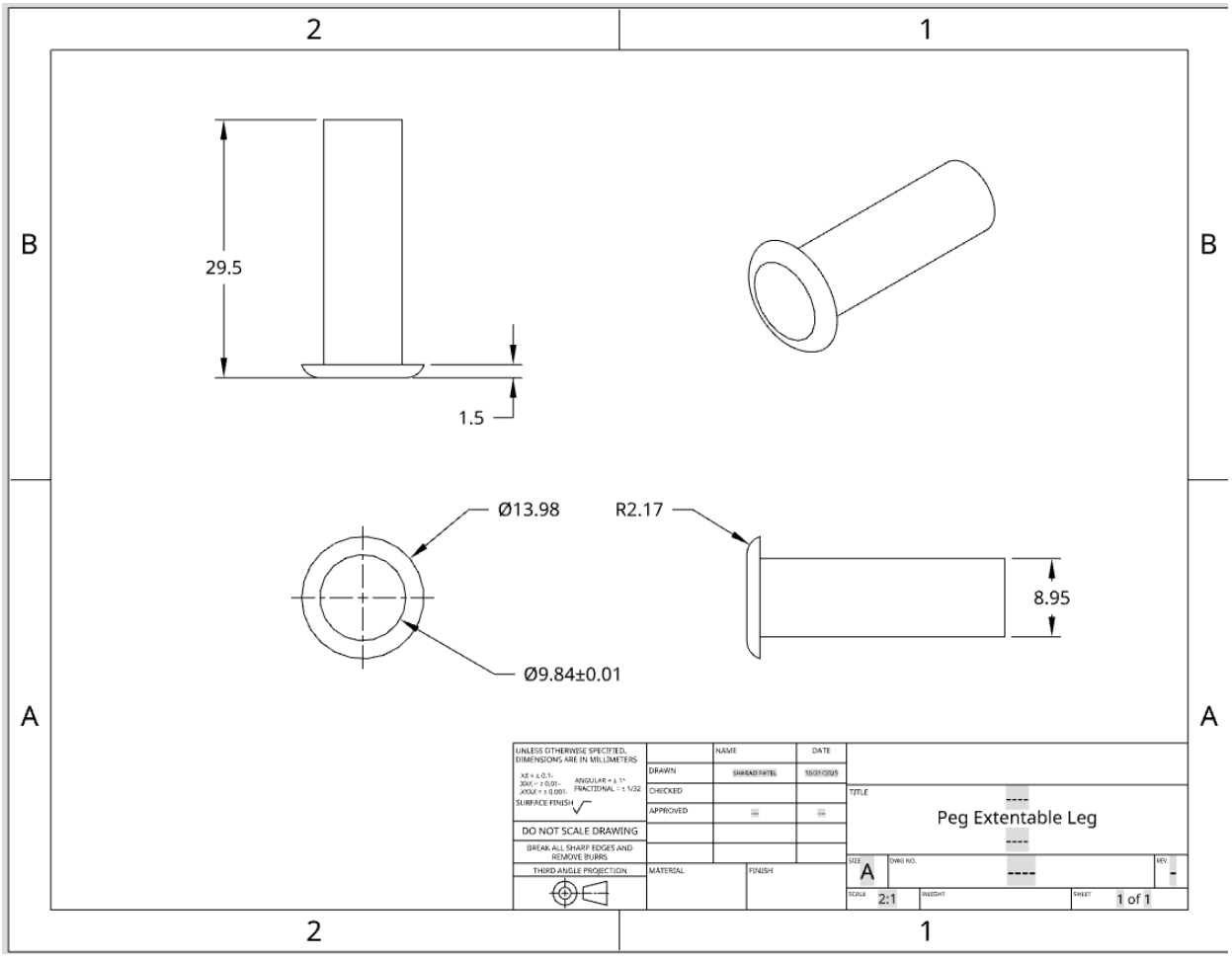


Figure 25: Multiview drawing of the Peg Extentable Leg

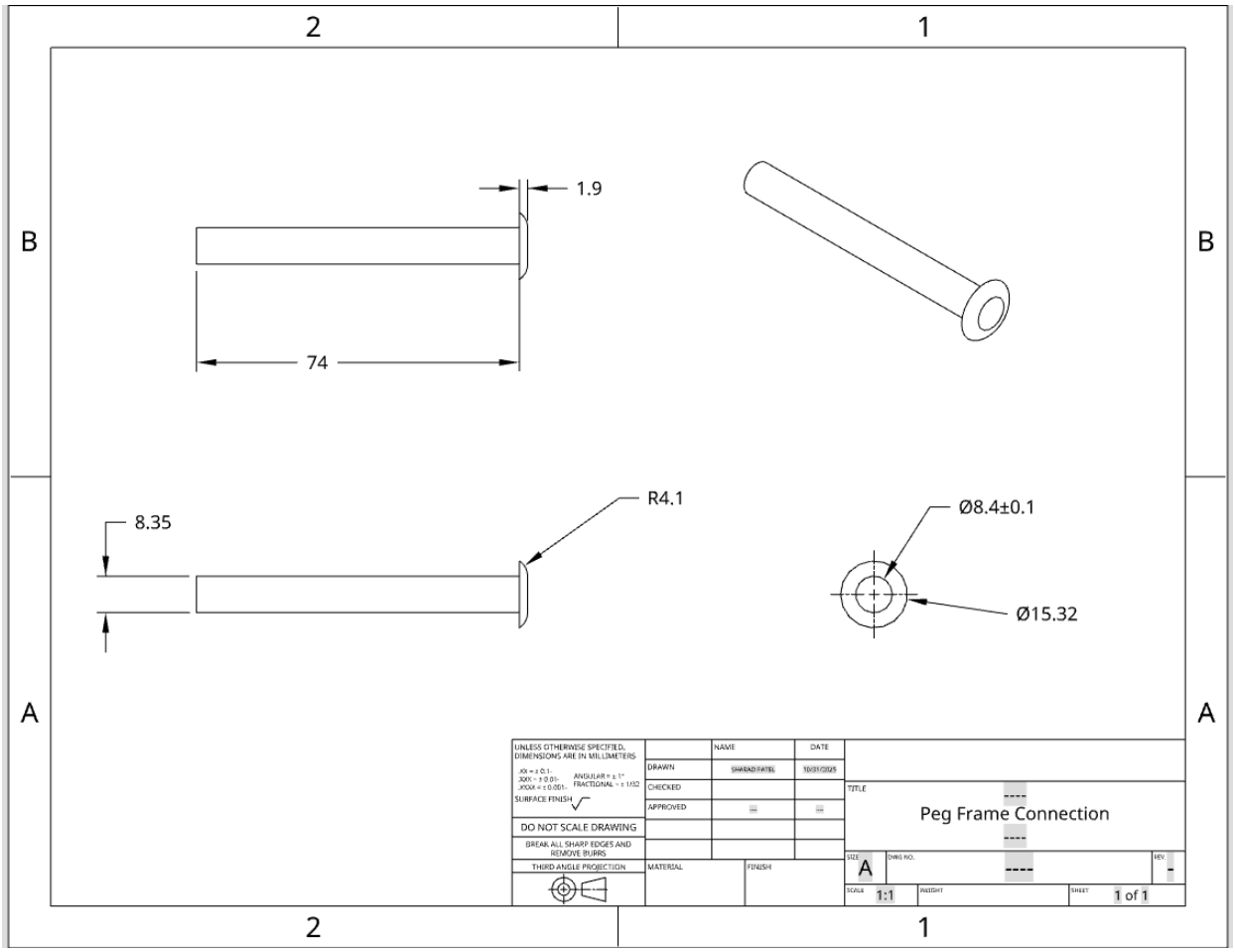


Figure 26: Multiview drawing of the Peg Frame Connector

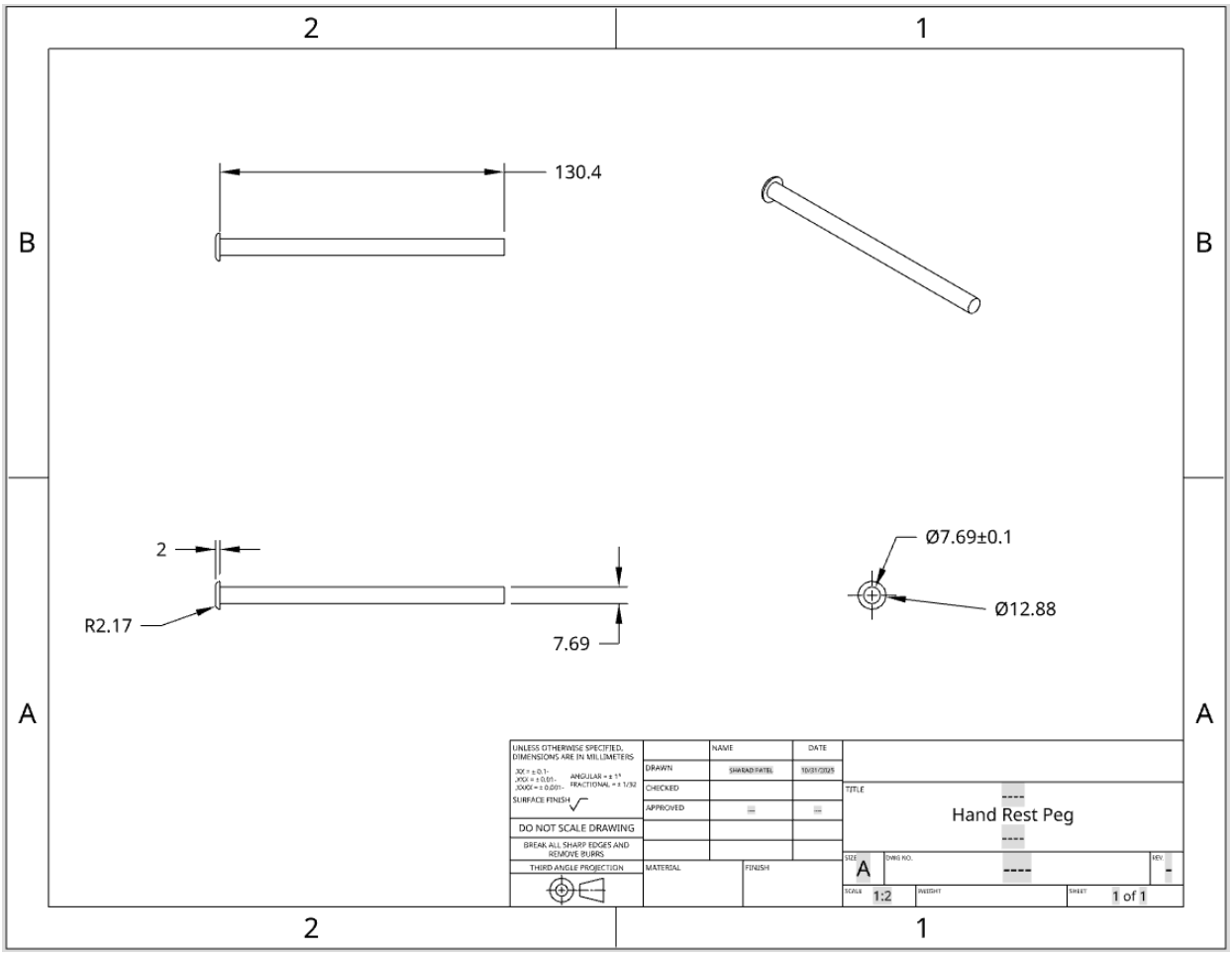


Figure 27: Multiview drawing of the Hand Rest Peg

Notes

These are my pegs, which are considered non unique parts, and thus grouped together. Overall, the best material for these will be stainless steel, to avoid rust and to have a lasting part with minimal wear. The general tolerances typically work for these pegs. I assigned the tolerance of the inner diameter of the peg to be 0.1 mm. This does not need to be too specific as the design has space built around the peg, and thus the fit isn't perfect as you can tighten the peg to ensure it sits well.

Figure Assembly

Components

The assembly has 6 unique parts, with 3 unique pegs as well.

- Left Upper Frame
- Lower Frame
- Frame Connector
- Extendable Leg
- Cap
- Shoulder Rest
- Hand Rest

And the 3 Pins:

- Peg Extendable
- Peg Frame Connection
- Peg Hand Connection

Mirrored components in the assembly are:

- Right Upper Frame
    - Mirrored of Left upper frame

Pictures

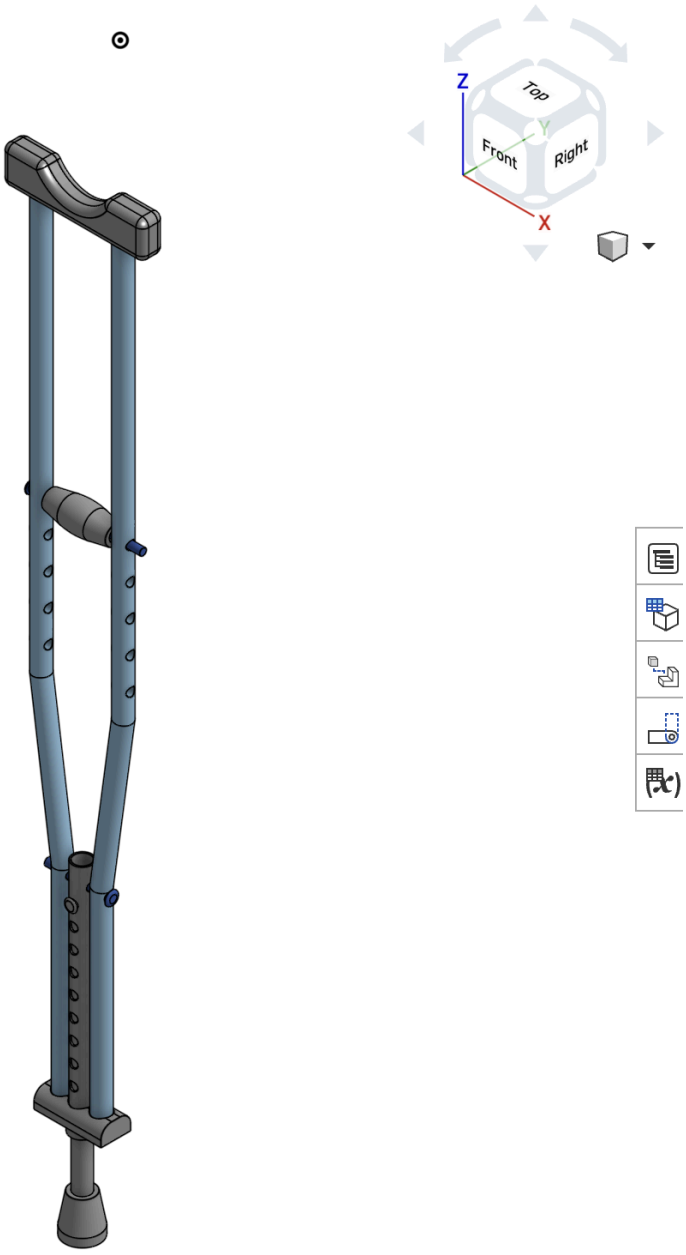


Figure 28 : Figure assembly from a standard isometric view. The figure is in its collapse state, it being the smallest it can be. The pins are also able to slide in and out. There is a limit to how much the pins can be pulled out in the direction of the head, with no limit on the other side.

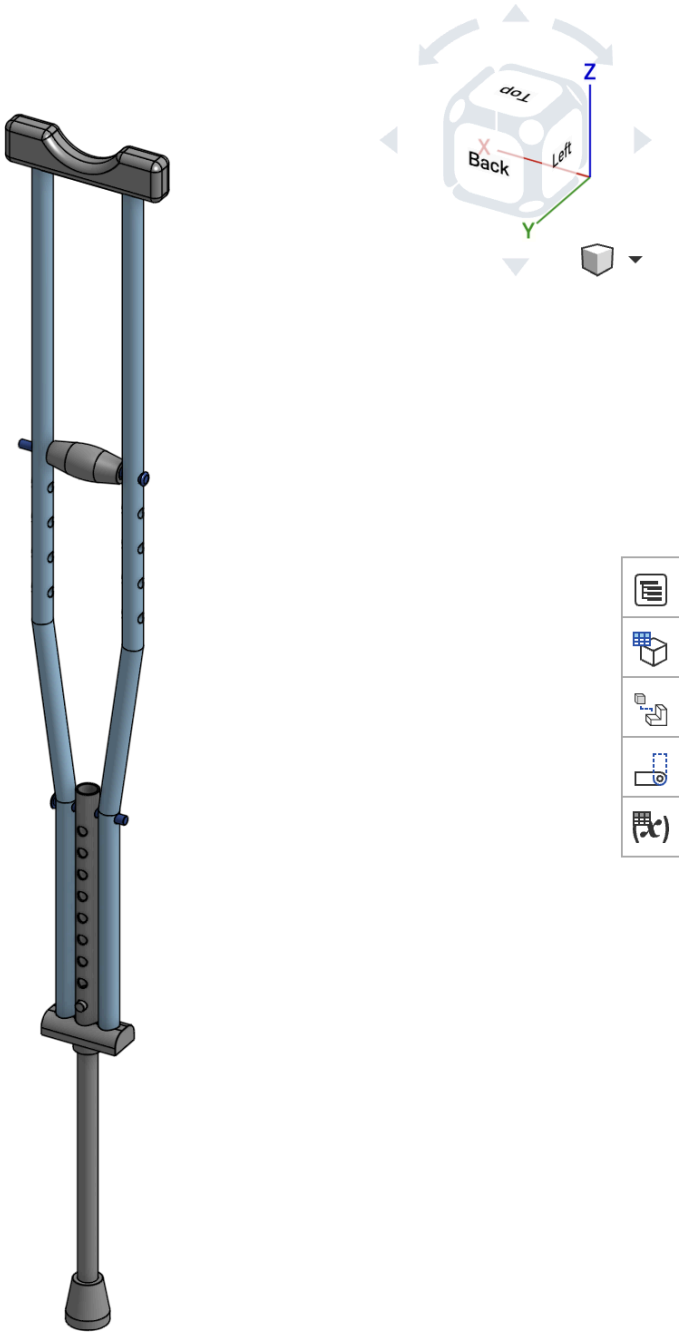


Figure 29 : Assembly as viewed from the back, left. This is the fully extended form of the figure.

Mated and Movement

- I used 2 mates for the entire assembly. Primarily, I used a fastened mate for all the parts fixed to each other. That includes the frames to the frame connector, the cap to the extendable leg, and the shoulder rest to the upper frames.
- I used the slider mate for the main motion of the assembly, which is the extendable leg that is telescopic to adjust for the height of the patient. There is a slider mate between

the top flat circular part of the extendable leg and the bottom flat circular part of the frame connector, and the direction of motion is up and down in the z axis. I set a limit from 47 to 230 mm so that the leg reaches all the pin holes in the lower frame and nothing more. There is also a slider mate between all the pins and their respective pin-holes so they can move in a and out, with a limit places so the head of the pin does not go through the solid frame.

Discussion

Strengths

- I think the strengths of my design is that it realistically incorporates modular, realistic crutch components that enable height adjustability. The telescoping mechanism allows for around 180 mm of additional height to cater to a greater range of subjects. Along with that, the forked frame with several pin hones allows for proper structural stability, providing the house for several comfort elements such as the shoulder and hand support, with the hand support being able to be adjusted via the pin holes.

Weaknesses

- The weakness of this design is that there are no features to fasten the pins onto the frame, and in the real world, it will slide out in a few uses. The hand and shoulder rests use a rudimentary design that is not optimized for comfort, and the expandable leg has a pin locking system rather than a traditional spring loaded locking system. Finally, we have not run simulations with materials to determine how load distribution will work with our design.

Future Directions

- For device to work properly, a few modifications that can be made are:
    - Adding a spring loaded system to secure the extendable leg
    - Run a load distribution simulation to determine structural strength under load
    - Add fasteners to the pins so the connection are secured

- Other changes I would make to my design are:
    - Improve the hand and shoulder supports to reduce pressure point.
    - Add a textured grip to the hand rest for better stability
    - Run a finite element analysis to minimize the weights while maintaining stiffness
    - Make the upper framed curved rather then having strength edges

Cad Link:
https://cad.onshape.com/documents/3332cd4350bcbad4a2f58dc2/w/0d04071f471f0c1d16dcb89a/e/5fd6f5a3437e5626d89ecf05?renderMode=0&uiState=69042282f2cf451dc6887ee3

References

1. Egbumike, C., & OBEAGU, P. R. O. F. E. M. M. A. N. U. E. L. (2021). Analysis of Method of Axillary Crutch Measurement. https://doi.org/10.9734/JAMMR/2021/XXXXX